

Report of Project 1

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1.Experimental Result:

Version	Description	Valid BPB	Valid Token_ppl	Test BPB
Baseline	LayerNorm + GELU + absolute positional embeddings	2.0711	79.5283	2.1013
B	RMSNorm only: all LayerNorm layers are replaced with RMSNorm; everything else remains the same as the baseline.	2.0779	80.6939	
C	RMSNorm + SwiGLU: RMSNorm normalization + SwiGLU feed-forward	2.0114	70.1036	
D	RMSNorm + SwiGLU + ALiBi: remove learnable positional embeddings and add ALiBi relative bias in attention	1.8618	51.1129	1.8911

Note: Models B and C were assessed solely on the validation set to support model selection. Only the optimal model D underwent a single test-set evaluation. This practice follows the requirement that the test set may only be used for final submission to prevent test-set leakage.

2.Reproducibility

2.1 Environment

Operating System: Windows, running in V.S Code terminal. Python version: 3.12. Deep learning framework: PyTorch 2.7.1. All experiments in this assignment run on CPU, with FP32 precision used for evaluation.

All the above items have passed the contract test tests/test_contract.py.

```
python -m unittest discover -s tests -v
```

2.2 Data and Code Description

The dataset used is the WikiText-2 corpus provided with the course materials, paired with the predefined BPE-2048 tokenizer. No modifications have been made to the data/ directory. Fixed project files including common.py, evaluate.py, and the baseline model.py remain untouched. All architectural modifications are implemented in student.py, and training configurations are adjusted within train.py. The model strictly follows the interface specification of the assignment. build_model(config) returns a Transformer model with a context length of 256. The forward pass outputs logits with shape [batch, time, 2048]. Causal attention is implemented to prevent access to future tokens, satisfying evaluation requirements.

2.3 Model Modifications

(1) RMSNorm (Version B): Compared with LayerNorm, RMSNorm removes mean centering and

only scales activations by their root mean square, leading to simpler computation. Theoretically, it improves training stability for small models.

- (2) SwiGLU Feed-forward Network (Version C, built upon RMSNorm) : The standard single-branch GELU MLP is replaced with SwiGLU. It performs element-wise multiplication between outputs of two separate linear projections, followed by an additional linear layer. This design yields stronger nonlinear expressiveness and can further reduce perplexity on many Transformer tasks.
- (3) ALiBi Relative Positional Bias (Version D, built upon Version C): Learnable absolute positional embeddings are removed. Instead, a negative bias linearly correlated with the relative token distance is added to attention logits. ALiBi directly encodes relative positions, reduces parameter count, preserves causal masking, and inherently achieves better length extrapolation.

#Fixed hyperparameters: width=128, heads=4, depth=4, context=256, vocab=2048.

2.4 Core change

All model modifications are implemented exclusively in student.py, while all other project files remain identical to the baseline. Three core components are implemented: RMSNorm, SwiGLU feed-forward module, and ALiBi causal attention. The code snippet is shown below:

```
class ALiBiAttention(nn.Module):
    def __init__(self, width=128, heads=4):
        super().__init__()
        self.heads = heads
        self.head_dim = width // heads
        self.qkv = nn.Linear(width, 3 * width)
        self.proj = nn.Linear(width, width)
        slopes = torch.tensor([2 ** (-8 / self.heads * (i+1)) for i in range(self.heads)])
        self.register_buffer("slopes", slopes)

    def forward(self, x):
        batch, length, width = x.shape
        qkv = self.qkv(x).view(batch, length, 3, self.heads, self.head_dim).permute(2, 0, 3, 1, 4)
        q, k, v = qkv[0], qkv[1], qkv[2]
        att_logits = (q @ k.transpose(-2, -1)) / (self.head_dim ** 0.5)
        pos = torch.arange(length, device=q.device).unsqueeze(-1)
        dist = pos.unsqueeze(0) - pos.unsqueeze(1) # shape [T, T]
        dist = dist.float()
        causal_mask = dist > 0
        dist = dist.masked_fill(causal_mask, float("-inf"))
        alibi_bias = dist * self.slopes.view(-1,1,1)
        att_logits = att_logits + alibi_bias
        att_weights = F.softmax(att_logits, dim=-1)
        attended = att_weights @ v
        attended = attended.transpose(1, 2).reshape(batch, length, width)
        return self.proj(attended)
```

During model construction: learnable absolute positional embeddings from the baseline are removed. All LayerNorm layers are replaced with RMSNorm, the original GELU MLP is substituted with SwiGLUFeedForward, and the attention block is replaced with ALiBiAttention.

2.5 Model Selection Pipeline

The final submitted version is Version D. The reasons are as follows: each ablation variant is trained separately with a fixed random seed: Seed=17, which generates checkpoint.pt and metrics.json. Valid BPB values are compared.

- (1)A-B (RMSNorm only): Valid BPB rises from 2.011 to 2.078. For this small model with width=128 and depth=4, replacing all LayerNorm with RMSNorm alone slightly degrades performance despite its simpler computation.
- (2)B-C (adding SwiGLU): BPB falls back from 2.078 to 2.011, matching the baseline level. The stronger nonlinear expressiveness of SwiGLU offsets the performance penalty introduced by RMSNorm alone.
- (3)C-D (adding ALiBi): Valid BPB drops substantially to 1.86186, and token PPL decreases significantly. Removing learnable positional embeddings and adopting relative positional bias in attention delivers the largest gain for this small model. The final frozen Model D achieves a Test BPB of 1.89110, the lowest among all four variants.
- (4)Ablation conclusion: RMSNorm alone slightly hurts performance. Combining RMSNorm with SwiGLU restores baseline performance. Introducing ALiBi on top of the RMSNorm+SwiGLU base is the most effective modification in this work; it reduces BPB while lowering the total number of parameters.

2.6Final test evaluation command

```
python evaluate.py --checkpoint runs/expD_alibi/checkpoint.pt --device cpu --precision fp32 --split test
```

3.Result

3.1 The final score

Test bpb=1.89110

Token_ppl=52.1039

3.2 Resource limited

Constraint	Experiment	status
CPU evaluation time $\leq 5 \times$ baseline evaluation time.	Baseline: 6.12s D:8.41s, $8.41 < 30.6$	PASS
Peak RAM ≤ 4 GiB	706.9MB	PASS
Inference asset size ≤ 64 MiB	4.02MB	PASS

Conclusion:

Full test evaluations for both the baseline model and frozen Model D were conducted on the same machine with identical CPU environment. Model D took 8.41 seconds for evaluation, while the baseline took 6.12 seconds. The runtime of Model D is below the upper bound of $5 \times$ the baseline runtime. The peak RAM usage of the Python process during evaluation was 706.9 MB, which is under the 4 GiB limit. The only inference asset is checkpoint.pt, with a file size of 4.02 MB, well below the uncompressed 64 MiB cap. All three resource constraints are satisfied, and supporting screenshots are attached hereafter.

3.3 Ablation conclusion summary:

ALiBi is the module contributing the most performance gain in this work. By injecting distance-decaying bias directly into attention logits to replace learnable absolute positional embeddings, the model better models relative dependencies between tokens without increasing the number of parameters, leading to substantial reductions in both validation and test-set BPB.

4. Conclusion

This assignment implements a small autoregressive Transformer language model trained from scratch. Four progressive ablation experiments are conducted to verify the contributions of three modifications: RMSNorm, SwiGLU and ALiBi. The final optimal Model D (RMSNorm + SwiGLU + ALiBi) achieves a Test BPB of 1.89110 on the full test set, the lowest among all four variants. All assignment requirements are fully satisfied throughout the work, including causal constraints, interface contract, preservation of untouched fixed files, single test-set evaluation, and compliance with the three resource limits.

Appendix

Baseline:

```
code > runs > baseline > {} test_cpu_fp32.json > ...
1  {
2    "bpb": 2.1012604380869244,
3    "token_ppl": 80.84711881780987,
4    "nll_nats": 1881794.64505621,
5    "targets": 428405,
6    "utf8_bytes": 1292013,
7    "seconds": 10.189078900031745,
8    "protocol": "7506-mp1-wt2-v2",
9    "split": "test",
10   "precision": "fp32",
11   "checkpoint_sha256": "b7fdb67c8ef14b5166c0de1f17cf306f34800b41dfe97da856c9812cdb1f9701",
12   "implementation_sha256": "6cce142a31d396f017319543f50d70f30fa6da740729af01c8baed13c6f671f9",
13   "evaluator_sha256": "128bcb2dab0be0d427505bddb4671e3ab3a8f78e114be79a689c0f9029af133d",
14   "tokenizer_sha256": "020d1bc6aa4449c4f352b2e03d0e0fb4f39287f15297705e421b1fa7d817262e",
15   "device": "cpu",
16   "device_name": "CPU",
17   "peak_allocated_gb": 0.0,
18   "peak_reserved_gb": 0.0
19 }
```

Version B:

```
code > runs > expB_only_rmsnorm > {} validation_cpu_fp32.json > ...
1  {
2    "bpb": 2.077969768176415,
3    "token_ppl": 80.69386455373758,
4    "nll_nats": 1653519.1236153299,
5    "targets": 376599,
6    "utf8_bytes": 1148007,
7    "seconds": 4.887922800146043,
8    "protocol": "7506-mp1-wt2-v2",
9    "split": "validation",
10   "precision": "fp32",
11   "checkpoint_sha256": "e696ed84dcdcb09f4d98ade366ea76d25c927a4579b0a3c96926468c91b4b978",
12   "implementation_sha256": "86ceca923c81ec1f52a25c4dc48973be32d517219fb664330db17ae2a7bb7aff",
13   "evaluator_sha256": "128bcb2dab0be0d427505bddb4671e3ab3a8f78e114be79a689c0f9029af133d",
14   "tokenizer_sha256": "020d1bc6aa4449c4f352b2e03d0e0fb4f39287f15297705e421b1fa7d817262e",
15   "device": "cpu",
16   "device_name": "CPU",
17   "peak_allocated_gb": 0.0,
18   "peak_reserved_gb": 0.0
19 }
```

Version C:

```
{
  "bpb": 2.0113865417047836,
  "token_ppl": 70.1036592532632,
  "nll_nats": 1600536.3324462979,
  "targets": 376599,
  "utf8_bytes": 1148007,
  "seconds": 5.131659399718046,
  "protocol": "7506-mp1-wt2-v2",
  "split": "validation",
  "precision": "fp32",
  "checkpoint_sha256": "b567688b6316c7dc216c8f3357d66243f7b2c0d98d2090ae201db1d3e28b7403",
  "implementation_sha256": "3ae6bbbf1d7d2848159535408692a3a6bd73f1170fa5409ed8ac8b585d3dcf3",
  "evaluator_sha256": "128bcb2dab0be0d427505bddb4671e3ab3a8f78e114be79a689c0f9029af133d",
  "tokenizer_sha256": "020d1bc6aa4449c4f352b2e03d0e0fb4f39287f15297705e421b1fa7d817262e",
  "device": "cpu",
  "device_name": "CPU",
  "peak_allocated_gb": 0.0,
  "peak_reserved_gb": 0.0
}
```

Version D:

```
code > runs > expD_alibi > {} validation_cpu_fp32.json > ...
1 {
2   "bpb": 1.861862225382973,
3   "token_ppl": 51.112874591865754,
4   "nll_nats": 1481554.1796401984,
5   "targets": 376599,
6   "utf8_bytes": 1148007,
7   "seconds": 9.556115100160241,
8   "protocol": "7506-mp1-wt2-v2",
9   "split": "validation",
10  "precision": "fp32",
11  "checkpoint_sha256": "12c60400024365e09aea8144ccb55d278531e88415aee945fe36f205d50f0195",
12  "implementation_sha256": "9a24a6251776fcdab4f27a4d7f07d47eed43abec7fcb658f9fdf6a92a635b5c5",
13  "evaluator_sha256": "128bcb2dab0be0d427505bddb4671e3ab3a8f78e114be79a689c0f9029af133d",
14  "tokenizer_sha256": "020d1bc6aa4449c4f352b2e03d0e0fb4f39287f15297705e421b1fa7d817262e",
15  "device": "cpu",
16  "device_name": "CPU",
17  "peak_allocated_gb": 0.0,
18  "peak_reserved_gb": 0.0
19 }
```

Test:

```
code > runs > expD_alibi > {} test_cpu_fp32.json > ...
1 {
2   "bpb": 1.8911039024430205,
3   "token_ppl": 52.103921412041224,
4   "nll_nats": 1693587.8734299776,
5   "targets": 428405,
6   "utf8_bytes": 1292013,
7   "seconds": 8.410269100219011,
8   "protocol": "7506-mp1-wt2-v2",
9   "split": "test",
10  "precision": "fp32",
11  "checkpoint_sha256": "12c60400024365e09aea8144ccb55d278531e88415aee945fe36f205d50f0195",
12  "implementation_sha256": "19ecea564a95b576dd2e296549be0ea0c94e52e00230861b3098f2a818d59ee1",
13  "evaluator_sha256": "128bcb2dab0be0d427505bddb4671e3ab3a8f78e114be79a689c0f9029af133d",
14  "tokenizer_sha256": "020d1bc6aa4449c4f352b2e03d0e0fb4f39287f15297705e421b1fa7d817262e",
15  "device": "cpu",
16  "device_name": "CPU",
17  "peak_allocated_gb": 0.0,
18  "peak_reserved_gb": 0.0
19 }
```

 checkpoint.pt

修改日期: 2026/9/28 1:40
大小: 4.02 MB

类型: PT 文件

大小: 4.02 MB

应用 (1)

 test_cpu_fp32.json - MP1_st...	0.2%	706.9 MB	0.1 MB/秒	0.1 Mbps
--	------	----------	----------	----------

```
OK  
(.venv) (base) PS E:\香港大学\7506\MP1_student_starter\code> python tests/test_contract.py  
.....  
-----  
Ran 5 tests in 0.051s  
OK
```